

CLASS 10 · CBSE · NCERT MATHEMATICS · SHORT NOTES

# Chapter 11: Areas Related to Circles

Topic-wise short notes with solved examples

A sector is a "slice" of a circle and a segment is the part cut off by a chord. This chapter is about finding their areas and the length of an arc.

## 1 Circle: a quick recap

- For a circle of radius  $r$ : circumference  $= 2\pi r$  and area  $= \pi r^2$ . Diameter  $d = 2r$ .
- Use  $\pi = \frac{22}{7}$  unless the question gives another value such as 3.14.
- Area of a semicircle  $= \frac{1}{2}\pi r^2$ . Area of a quadrant  $= \frac{1}{4}\pi r^2$ .
- Area of a ring between two concentric circles of radii  $R$  and  $r$  is  $\pi(R^2 - r^2)$ .

## 2 Sector of a circle

- A **sector** is the region enclosed by two radii and the arc between them.
- The angle between the two radii is the **angle of the sector**,  $\theta$ .
- The smaller region is the **minor sector** (angle  $\theta$ ) and the larger is the **major sector** (angle  $360^\circ - \theta$ ).
- The whole circle is a sector of angle  $360^\circ$ , so a sector of angle  $\theta$  is the fraction  $\frac{\theta}{360}$  of the circle.

$$\text{Area of sector} = \frac{\theta}{360} \times \pi r^2 \quad \text{Length of arc} = \frac{\theta}{360} \times 2\pi r$$

- Perimeter of a sector = length of arc +  $2r$ .
- Area of the major sector  $= \pi r^2 - \text{area of the minor sector}$ .

**Example 2.1** Find the area of a sector of a circle of radius 7 cm with angle  $60^\circ$ , and the length of its arc.

SOLUTION

$$\text{Area} = \frac{60}{360} \times \frac{22}{7} \times 7 \times 7 = \frac{1}{6} \times 154 = \frac{77}{3} \text{ cm}^2$$

$$\text{Arc} = \frac{60}{360} \times 2 \times \frac{22}{7} \times 7 = \frac{1}{6} \times 44 = \frac{22}{3} \text{ cm}$$

**Area  $\approx 25.67 \text{ cm}^2$ ; arc length  $\approx 7.33 \text{ cm}$**

**Example 2.2** The minute hand of a clock is 21 cm long. Find the area it sweeps in 10 minutes.

**SOLUTION**

In 60 minutes the minute hand turns  $360^\circ$ , so in 10 minutes it turns  $60^\circ$ .

$$\text{Area} = \frac{60}{360} \times \frac{22}{7} \times 21 \times 21 = \frac{1}{6} \times 1386$$

$$\text{Area} = 231 \text{ cm}^2$$

**Example 2.3** A horse is tied to a peg at one corner of a square field of side 15 m by a rope 7 m long. Find the area of the part of the field in which it can graze.

**SOLUTION**

The corner of a square is  $90^\circ$ , so the horse grazes a sector of radius 7 m and angle  $90^\circ$ .

$$\text{Area} = \frac{90}{360} \times \frac{22}{7} \times 7 \times 7 = \frac{1}{4} \times 154$$

$$\text{Area} = 38.5 \text{ m}^2$$

**REMEMBER**

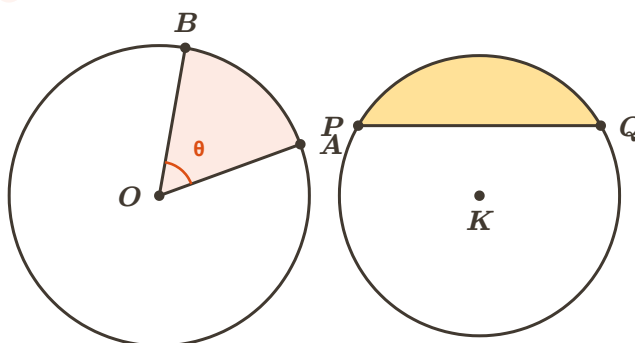
The minute hand turns  $6^\circ$  per minute. The hour hand turns  $30^\circ$  per hour, which is  $0.5^\circ$  per minute.

### 3 Segment of a circle

- A **segment** is the region between a chord and the arc it cuts off.
- The smaller region is the **minor segment** and the larger is the **major segment**.

Area of minor segment = area of the sector – area of the triangle formed by the chord and the two radii

- Area of the major segment =  $\pi r^2$  – area of the minor segment.

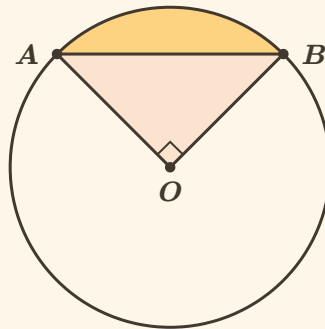


Left: sector OAB. Right: the minor segment cut off by the chord PQ.

- The triangle is isosceles with two sides equal to  $r$  and the angle  $\theta$  between them.

| Angle $\theta$ | Area of the triangle                           |
|----------------|------------------------------------------------|
| $60^\circ$     | $\frac{\sqrt{3}}{4}r^2$ (equilateral triangle) |
| $90^\circ$     | $\frac{1}{2}r^2$ (right-angled triangle)       |
| $120^\circ$    | $\frac{\sqrt{3}}{4}r^2$                        |

**Example 3.1** A chord of a circle of radius 14 cm subtends a right angle at the centre. Find the area of the minor segment and of the major segment.



Minor segment (yellow) = sector OAB – triangle OAB

SOLUTION

$$\text{Area of sector} = \frac{90}{360} \times \frac{22}{7} \times 14 \times 14 = 154 \text{ cm}^2$$

$$\text{Area of triangle} = \frac{1}{2} \times 14 \times 14 = 98 \text{ cm}^2$$

$$\text{Minor segment} = 154 - 98 = 56 \text{ cm}^2$$

$$\text{Area of circle} = \frac{22}{7} \times 14 \times 14 = 616 \text{ cm}^2, \text{ so major segment} = 616 - 56 = 560 \text{ cm}^2$$

**Minor segment = 56 cm<sup>2</sup>; major segment = 560 cm<sup>2</sup>**

**Example 3.2** A chord of a circle of radius 21 cm subtends an angle of  $60^\circ$  at the centre. Find the area of the minor segment. Take  $\sqrt{3} = 1.732$ .

SOLUTION

$$\text{Area of sector} = \frac{60}{360} \times \frac{22}{7} \times 21 \times 21 = 231 \text{ cm}^2$$

$$\text{The triangle is equilateral with side 21 cm: area} = \frac{\sqrt{3}}{4} \times 21 \times 21 = 110.25 \times 1.732 \approx 190.95 \text{ cm}^2$$

**Area of the minor segment  $\approx 231 - 190.95 = 40.05 \text{ cm}^2$**

#### 4 Finding the area of the triangle for other angles

- Draw the perpendicular  $OM$  from the centre to the chord  $AB$ . It bisects the chord and the angle  $\theta$ .
- $OM = r \cos \frac{\theta}{2}$  and  $AM = r \sin \frac{\theta}{2}$ , so the chord  $AB = 2r \sin \frac{\theta}{2}$ .

- Area of triangle =  $\frac{1}{2} \times AB \times OM$ .

**Example 4.1** Find the area of the triangle formed by two radii and a chord that subtends  $120^\circ$  at the centre of a circle of radius 12 cm.

SOLUTION

$OM = 12 \cos 60^\circ = 6$  cm and  $AM = 12 \sin 60^\circ = 6\sqrt{3}$  cm, so  $AB = 12\sqrt{3}$  cm

$$\text{Area} = \frac{1}{2} \times 12\sqrt{3} \times 6$$

$$\text{Area} = 36\sqrt{3} \approx 62.35 \text{ cm}^2$$

## 5 Common mistakes

- Using the diameter in place of the radius.
- Forgetting to add  $2r$  when the **perimeter** of a sector is asked.
- Giving the sector's area when the segment's area is asked.
- Using  $\frac{22}{7}$  when the question says  $\pi = 3.14$ , or the other way round.
- Leaving out the units: cm for length,  $\text{cm}^2$  for area.